How does Saturated Fat Impairs Circadian Transciptomics?

BCB420 Analysis Presentation

Zhanyan Guo

GEO dataset GSE205677

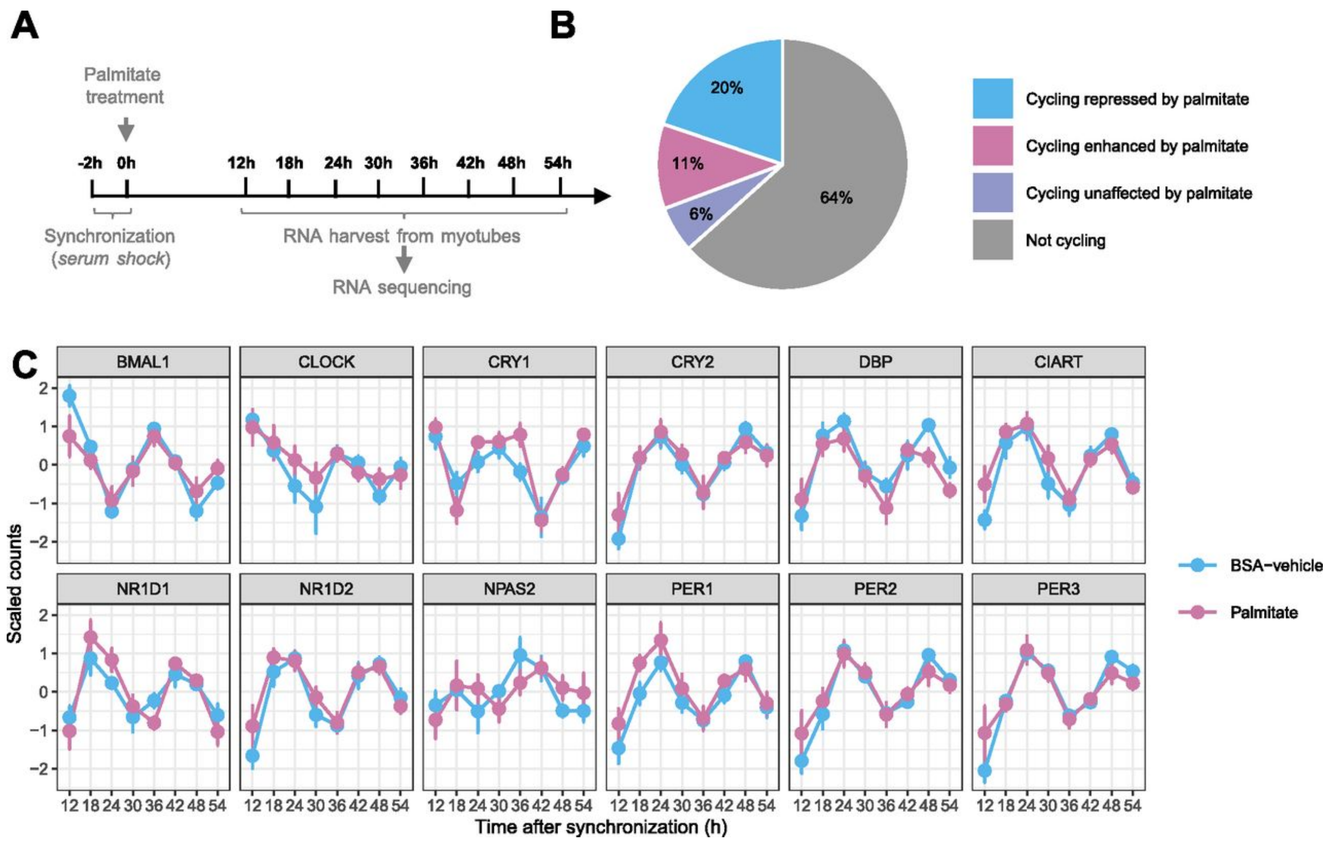
Palmitate treatment vs Control

Primary skeletal muscle cells were synchronized, treated with palmitate, and harvested every 6 h

Each time point:

5~7 repeated samples per Group

Total: 107 samples



Palmitate alters the pattern of rhythmic transcripts.
(A) Graphic representation of data collection and RNA-seq of synchronized primary human skeletal muscle myotubes (n = 7) treated with palmitate (0.4 mM) or BSA-vehicle. (B) Proportion of rhythmic genes (RAIN analysis, FDR < 0.1) cycling only in the BSA condition (repressed by palmitate), only in the palmitate condition (enhanced by palmitate), or in both conditions (unaffected by palmitate). (C) Core clock gene expression in synchronized myotubes. Data are the mean ± SE, n = 7. (1)

- Citation:. Lipids impair circadian transcriptomics

Nicolas J Pillon, Laura Sardón Puig, Ali Altıntaş, Prasad G Kamble, Salvador Casaní-Galdón, Brendan M Gabriel, Romain Barrès, Ana Conesa, Alexander V Chibalin, Erik Näslund, Anna Krook, Juleen R Zierath
Life Science Alliance Oct 2022, 6 (1) e202201598; DOI: 10.26508/lsa.202201598

Effect on Expression Level or Cyclic Behaviour?

Original paper:
Use RAIN to identify cyclic Gene sets

Perform differential gene expression on selected Gene

Perform ORA on selected Gene

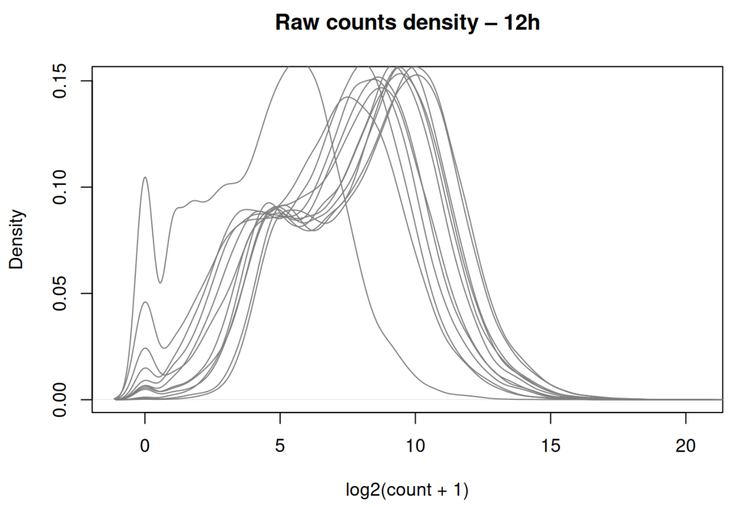
This Analysis:
Use timepoint + treatment design

Perform differential gene expression on full Gene Set

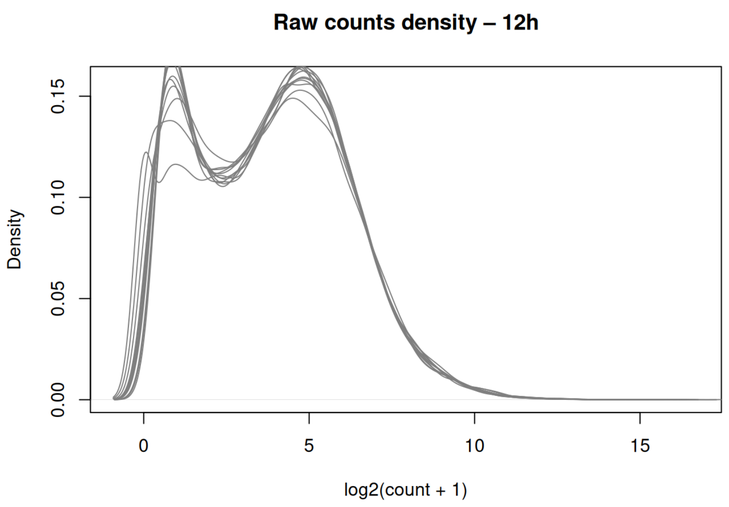
Perform ORA on full Gene Set

Normalization and Scoring

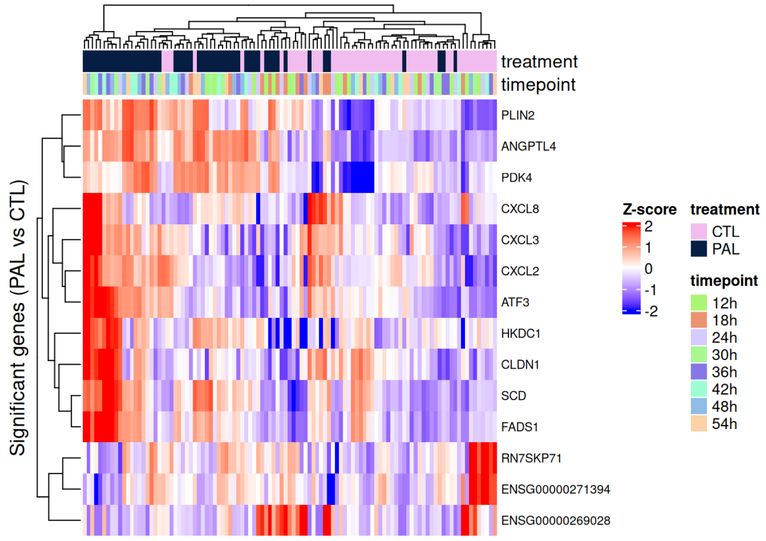
- Filter by 10 Gene count per (Treatment + Timepoint) Group
- Normalize using TMM
- Score with edgeR quasi-likelihood testing
- Raw **P value < 0.05**: 1223 Genes **FDR 0.05, |log (fold change)| > 1:** 14 Genes



Before Normalization



After Normalization



Heatmap of 14 Genes group by treatment

GSEA PreRanked (v4.4.0) Methods

Input

- Ranked gene list from edgeR
- Ranking score:

$$\mathrm{sign}(\mathrm{log2FC}) \times -\log_{10}(p\text{-value})$$

Key Parameters

- Permutations: 1000
- Scoring: weighted
- Gene set size: 15–200
- Seed: 12345

Gene Sets

- Bader Lab GMT (Sep 1, 2025 release)
- *Human_GOBP_AllPathways_noPFOCR_no_GO_iea*
- GO Biological Process + curated pathways
- Excluded IEA (high-confidence annotations)

- Use Raw P value: GSEA has built in Multiple hypothesis Correction
- Full Gene list: detect coordinated pathway shifts, even when individual genes are weak

References:
Subramanian et al., 2005 (GSEA); Bader Lab, 2025 (Gene sets)

Results by Enrichment Score ranking

Among the **positivel**y enriched pathways were processes related to:

- mTORC1 signaling (metabolic Stress)
- regulation of cholesterol biosynthesis
- lipid biosynthetic processes
- PPAR-α regulated lipid metabolism
- steroid metabolism
- endoplasmic reticulum (ER) stress response

mitochondrial respiration and oxidative phosphorylation were **negatively** enriched

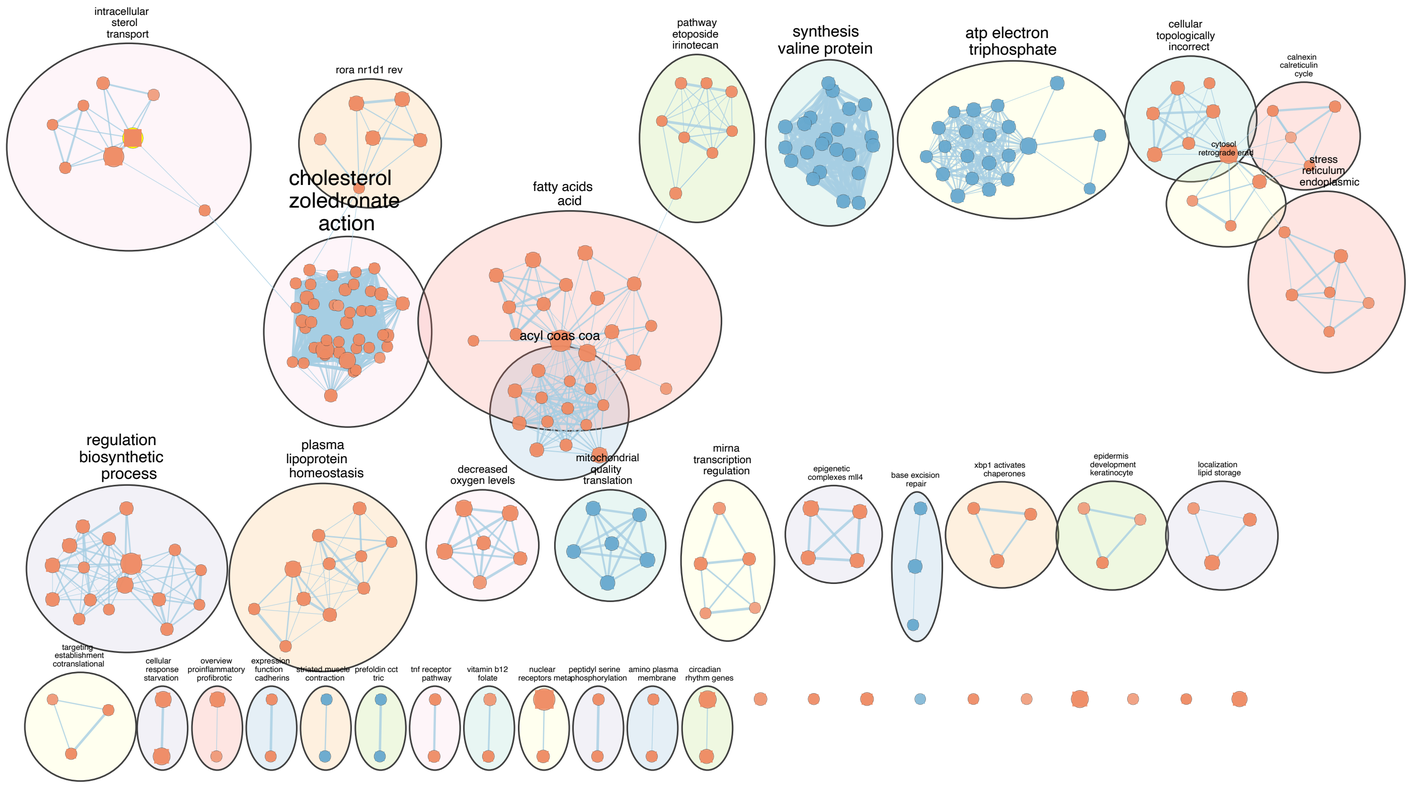
Top 10 Positively enriched Geneset

	GS follow link to MSigDB
1	HALLMARK_MTORC1_SIGNALING%MSIGDBHALLMARK%HALLMARK_MTORC1_SIGNALING
2	REGULATION OF RESPONSE TO ENDOPLASMIC RETICULUM STRESS%GOBP%GO:1905897
3	REGULATION OF CHOLESTEROL BIOSYNTHESIS BY SREBP (SREBF)%REACTOME DATABASE ID RELEASE 93%1655829
4	METABOLISM OF STEROIDS%REACTOME%R-HSA-8957322.4
5	CHOLESTEROL METABOLISM WITH BLOCH AND KANDUTSCH RUSSELL PATHWAYS%WIKIPATHWAYS_20250810%WP4718%HOMO SAPIENS
6	NEGATIVE REGULATION OF RESPONSE TO ENDOPLASMIC RETICULUM STRESS%GOBP%GO:1903573
7	REGULATION OF LIPID BIOSYNTHETIC PROCESS%GOBP%GO:0046890
8	REGULATION OF LIPID METABOLISM BY PPARALPHA%REACTOME%R-HSA-400206.8
9	EPIGENETIC REGULATION OF GENE EXPRESSION BY MLL3 AND MLL4 COMPLEXES%REACTOME%R-HSA-9818564.1
10	EPIGENETIC REGULATION OF ADIPOGENESIS GENES BY MLL3 AND MLL4 COMPLEXES%REACTOME DATABASE ID RELEASE 93%9851695

Top 10 Negatively enriched Geneset

	GS follow link to MSigDB
1	ELECTRON TRANSPORT CHAIN OXPHOS SYSTEM IN MITOCHONDRIA%WIKIPATHWAYS_20250810%WP111%HOMO SAPIENS
2	OXIDATIVE PHOSPHORYLATION%WIKIPATHWAYS_20250810%WP623%HOMO SAPIENS
3	OXIDATIVE PHOSPHORYLATION%GOBP%GO:0006119
4	MITOCHONDRIAL COMPLEX I ASSEMBLY MODEL OXPHOS SYSTEM%WIKIPATHWAYS_20250810%WP4324%HOMO SAPIENS
5	AEROBIC ELECTRON TRANSPORT CHAIN%GOBP%GO:0019646
6	MITOCHONDRIAL ATP SYNTHESIS COUPLED ELECTRON TRANSPORT%GOBP%GO:0042775
7	MITOCHONDRIAL TRANSLATION ELONGATION%REACTOME%R-HSA-5389840.3
8	ATP SYNTHESIS COUPLED ELECTRON TRANSPORT%GOBP%GO:0042773
9	STRIATED MUSCLE CONTRACTION%REACTOME DATABASE ID RELEASE 93%390522
10	PROTON MOTIVE FORCE-DRIVEN ATP SYNTHESIS%GOBP%GO:0015986

Cytoscape Visualization Results

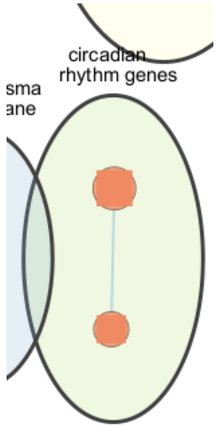


The EnrichmentMap was generated using the following thresholds:

- **FDR q-value cutoff:** 0.05
- **Edge similarity cutoff:** 0.375
- **Similarity metric:** Jaccard + overlap coefficient (default EnrichmentMap metric)
- **Gene set size limits:** 15–200 genes

Under these thresholds, the resulting network contained:

- **Nodes (gene sets):** *293*
- **Edges (similarity relationships):** 1594



- Major connections are Metabolic Rewiring and Lipid Metabolism
- Two Nodes of Circadian rhythmic Gene

Compare to Thresholded ORA (gprofiler2)

Input

- Significant DE genes from edgeR
  - FDR < 0.05
  - |log2FC| ≥ 1
- Analyzed separately:
  - up-regulated
  - down-regulated
  - combined

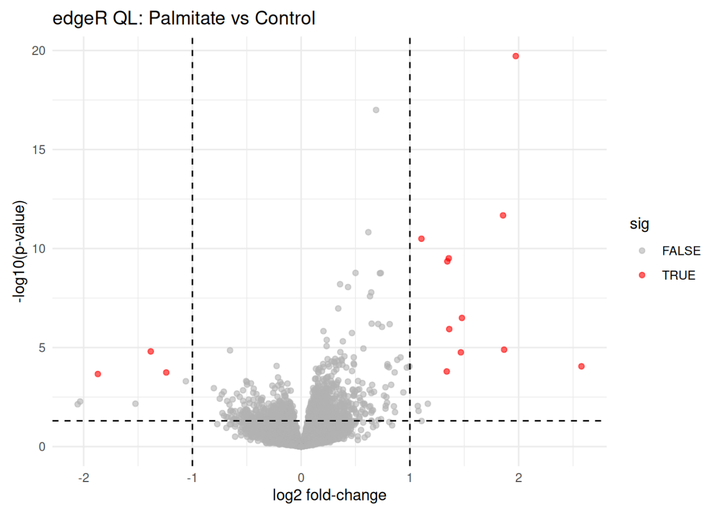
- 11 up-regulated
- 3 down-regulated

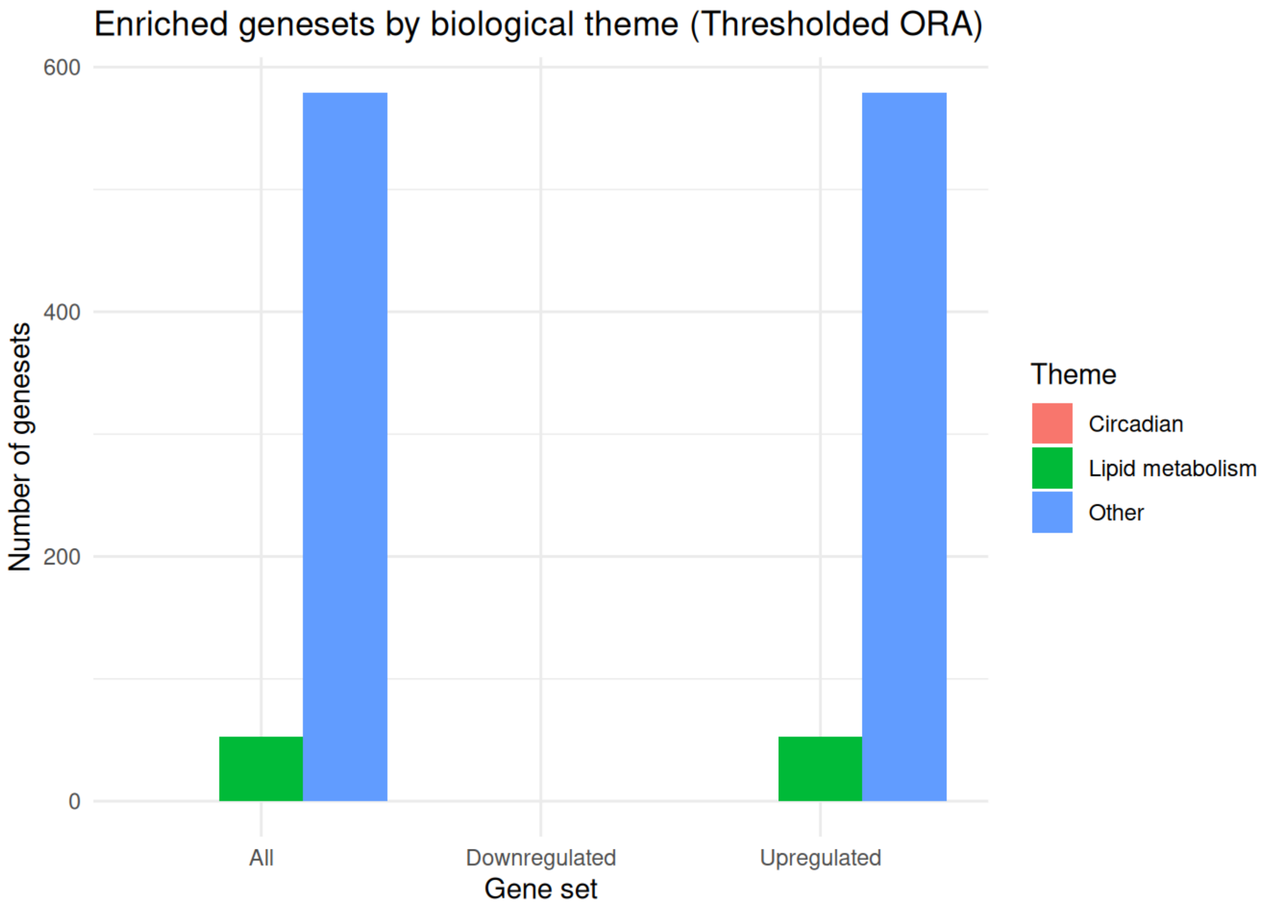
Gene Set Sources

- GO (BP, MF, CC)
- Reactome, KEGG, WikiPathways
- TF, miRNA
- CORUM, HPA, HP

Key Parameters

- `organism = "hsapiens"`
- `ordered_query = FALSE` (unranked gene list)
- `significant = FALSE` (return all results)
- `exclude_iea = TRUE` (remove low-confidence annotations)
- `correction_method = "fdr"`



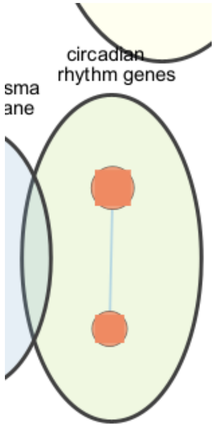


Circadian Gene Sets Not identified by Gprofiler

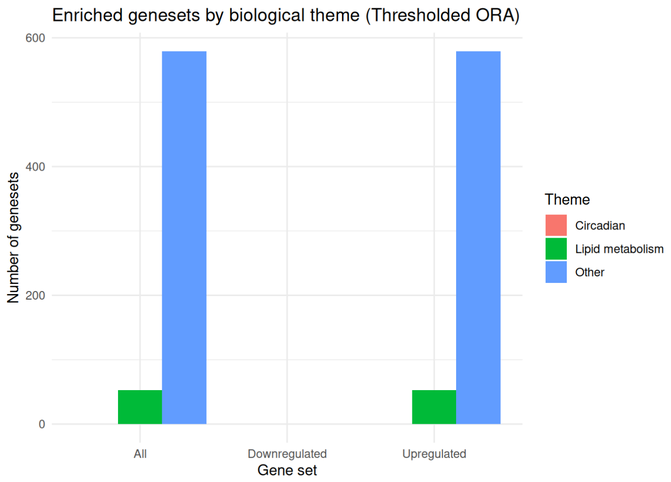
Collapsed Network has few Connected Themes

Both Analysis Shows Circadian Rhythmic Genes are Weakly Effected

- Original Paper use RAIN and identified: “Core clock genes, including *BMAL1*, *CIART*, *DBP*, *CRY1*, *CRY2*, *NR1D1*, *NR1 D2*, *PER1*, *PER2*, and *PER3*, were unaffected by palmitate “



- Consistent with less significant enrichment Results for Circadian Rhythmic Genes in this assignment



Unable to Differentiate Rhythmic Behaviour

- Our Results identifies only expression change that are well documented on lipid metabolic pathways (Georgiadi & Kersten, 2012)
- The original paper use RAIN to identify Rhythmic Expression change has small overlap with canonical lipid metabolic pathways
- Differential Expression has shortage on time dependent data

Both Study Shows enrichment in histone H3 modification Pathways

MLL3 / MLL4 are histone methyltransferases

They add **H3K4me1** marks at enhancer regions

- activates enhancers
- opens chromatin
- allows transcription factors to bind

Top 10 Positively enriched Geneset

	GS follow link to MSigDB
1	HALLMARK_MTORC1_SIGNALING%MSIGDBHALLMARK%HALLMARK_MTORC1_SIGNALING
2	REGULATION OF RESPONSE TO ENDOPLASMIC RETICULUM STRESS%GOBP%GO:1905897
3	REGULATION OF CHOLESTEROL BIOSYNTHESIS BY SREBP (SREBF)%REACTOME DATABASE ID RELEASE 93%1655829
4	METABOLISM OF STEROIDS%REACTOME%R-HSA-8957322.4
5	CHOLESTEROL METABOLISM WITH BLOCH AND KANDUTSCH RUSSELL PATHWAYS%WIKIPATHWAYS_20250810%WP4718%HOMO SAPIENS
6	NEGATIVE REGULATION OF RESPONSE TO ENDOPLASMIC RETICULUM STRESS%GOBP%GO:1903573
7	REGULATION OF LIPID BIOSYNTHETIC PROCESS%GOBP%GO:0046890
8	REGULATION OF LIPID METABOLISM BY PPARALPHA%REACTOME%R-HSA-400206.8
9	EPIGENETIC REGULATION OF GENE EXPRESSION BY MLL3 AND MLL4 COMPLEXES%REACTOME%R-HSA-9818564.1
10	EPIGENETIC REGULATION OF ADIPOGENESIS GENES BY MLL3 AND MLL4 COMPLEXES%REACTOME DATABASE ID RELEASE 93%9851695

(Lee et al., 2013; Herz et al., 2012), MLL4 is a critical epigenetic regulator of enhancer activity, Enhancer-associated histone modifications are mediated by MLL3 and MLL4

Original Study Suggests Genome Wide Change for Down Stream H3K27ac

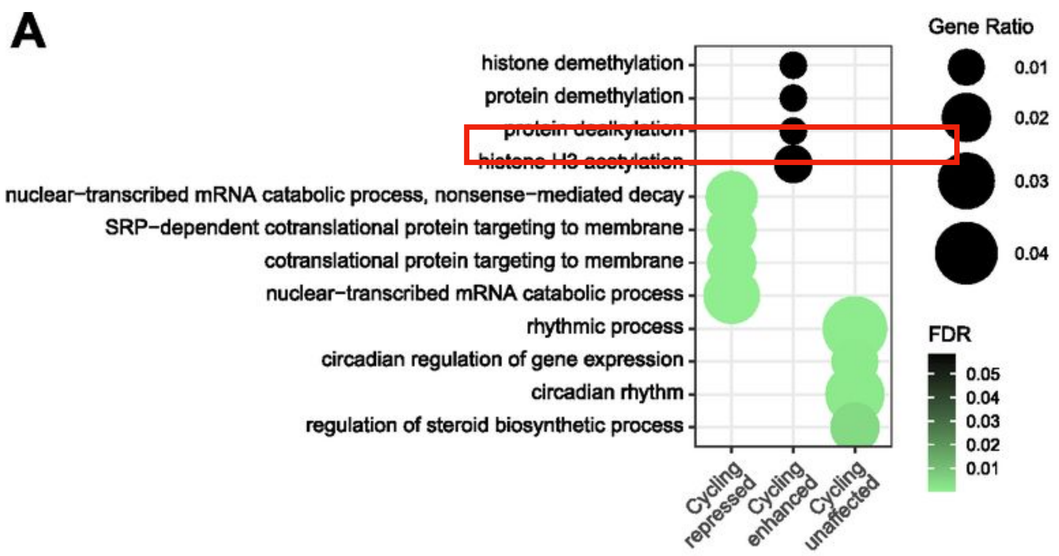
In the downstream

H3K4me1 marks recruits CBP/p300

adds **H3K27ac**

leads to increased gene expression

Original Study Reports histone H3 acetylation Cycling Enhanced

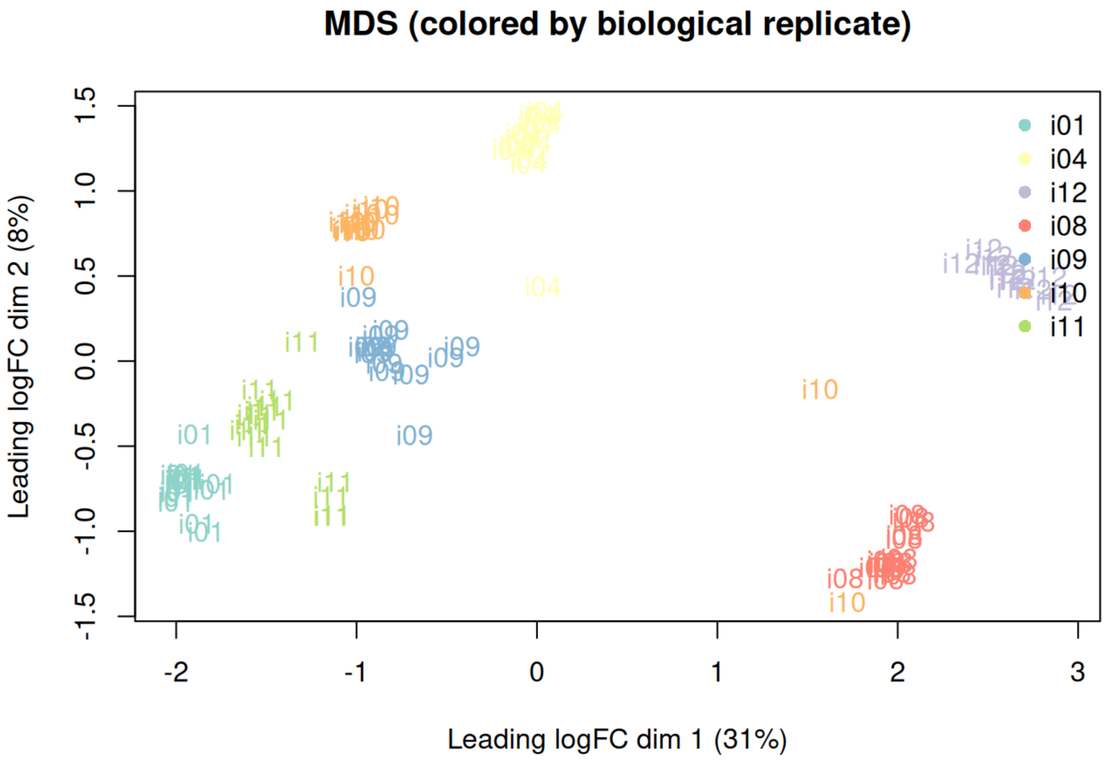


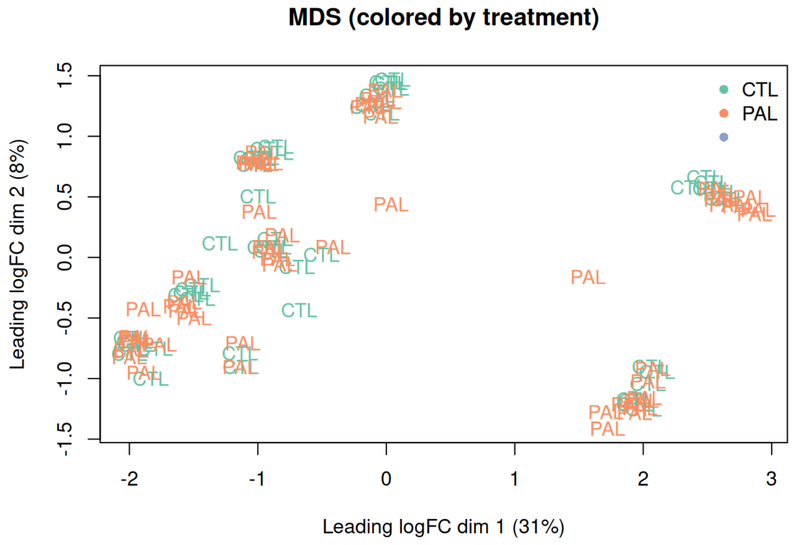
The original study performs further experiment to identify Palmitate exposure leads to **genome-wide changes in H3K27ac**

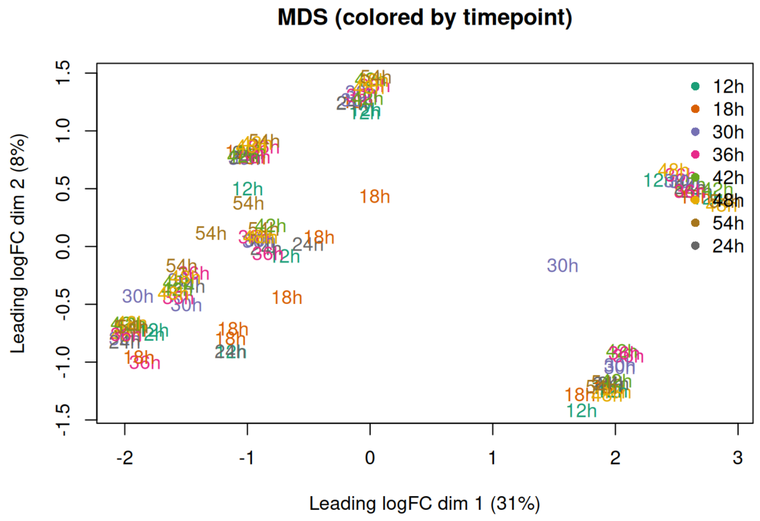
Shows circadian disruption happens through **metabolic** and **epigenetic** reprogramming

Discussion

Individual difference dominates?







References

Subramanian A. et al., 2005. *PNAS* — Gene Set Enrichment Analysis

Bader Lab, 2025. Gene Set Collections (GMT, Sept 1, 2025)

Herz H.M. et al., 2012. *Nature* — MLL3/MLL4 and enhancer regulation

Lee J.E. et al., 2013. *Nature Communications* — MLL4 and enhancer activity

Creyghton M.P. et al., 2010. *PNAS* — H3K27ac marks active enhancers